

Ch1) SDLC - Systems Development Life Cycle.

- The systems development life cycle is a phased approach to solving business problems.
- Developed through the use of a specific cycle of analyst and user activities.
- Each phase has unique user activities.

The seven Phases of the systems Development Life Cycle.

1. Identifying problems, opportunities, and objectives; 2. Determining human information requirements; 3. Analyzing system needs; 4. Designing the recommended system; 5. Developing and documenting software; 6. Testing and maintaining the system; 7. Implementing and evaluating the system.

The agile approach.

The agile approach is a software development approach based on Values, Principles, Core practices. The 4 values are communication, simplicity, feedback, courage. Agile approach is Interactive, Incremental. Frequent iterations are essential for successful system development.

The five stages of the agile modelling development process. 1. Exploration, 2. Planning, 3. Iterations to the first release; 4. Productionizing; 5. Maintenance.

Object-Oriented Systems Analysis and Design

- Alternate approach to the structured approach of the SDLC that is intended to facilitate the development of systems that must change rapidly in response to dynamic business environments.
 - Use unified modelling language (UML) to model object-oriented systems.
 - Each object is a computer representation of some actual thing or event.
- Generally works well in situations where complicated information systems are undergoing continuous maintenance, adaptation, and redesign.

Ch4) Information Gathering — Interactive Methods

Interviewing, Joint Application Design (JAD), Questionnaires

- Interviewing: - Interview preparation, - Question types, - Arranging questions, - The interview report.

User stories

- Joint Application Design (JAD): - Involvement, - Location

- Questionnaires: - Writing questions, - Using scales, - Design, - Administering

① Interviewing is an important method for collect data on human and system information requirements.

Interviews reveal information about: - Interviewee opinions, - Interviewee feelings, - Goals, - Key HCI concerns

Question types: Open-ended, Closed

② JAD can replace a series of interviews with the user community. JAD is a technique that allows the analyst to accomplish requirements analysis and design the user interface with the users in a group setting.

③ Questionnaires are useful in gathering information from key organization members about:

- Attitudes — what people in the organization say they want.
- Beliefs — what people think is actually true.
- Behavior — what organizational members do.
- Characteristic — properties of people or things.

④ stories.

- stories originate in the workplace
- Organizational stories are used to relay some kind of information.
- When a story is told and retold over time it takes a mythic quality.
- Isolated stories are good when you are looking for facts.
- Enduring stories capture all aspects of the organization and are the ones a system analyst should look be seeking.

Ch5) Information Gathering — Unobtrusive Methods.

Unobtrusive Methods.

- Less disruptive
- Text analytics to analyze qualitative data
- Insufficient when used alone
- Multiple methods approach
- Used in conjunction with interactive methods

① Sampling: A process of systematically selecting representative elements of a population.

Involves two key decisions, what to examine, which people to consider.

Investigation: The act of discovery and analysis of data.

Hard data, - Quantitative, - Qualitative

② Analyzing Quantitative Documents.

- Reporting used for decision making
- Performance reports
- Records
- Data capture forms
- Ecommerce and other transactions

③ Analyzing Qualitative Documents

- Email messages
- Memos
- Signs of posters on bulletin boards
- Corporate Web sites (note the interactivity of Web sites)
- Manuals
- Policy handbooks.

④ Text Analytics.

- Software that can analyze unstructured qualitative data from any source including: 1. Transcripts, 2. Written reports, 3. Customers' communication collected through email, wikis, chat rooms, and other social networking sites.

⑤ Observation:

- Observation provides insight on what organizational members actually do.
- See firsthand the relationships that exist between decision-makers and other organizational members.
- Can also reveal important clues regarding HCI concerns.

⑥ STROBE:

Structured Observation of the Environment — a technique for observing the decision-maker's physical environment.

- Often it is possible to observe the particulars of the organizational surroundings that will confirm or negate the organizational narrative
- Also called stories or dialogue
- Information that is found through interviews or questionnaires.

⑦ Apply STROBE:

- The five symbols used to evaluate how observation of the elements of STROBE compared with interview results, are:
- A checkmark means the narrative is confirmed
- A "X" means the narrative is ~~confirmed~~ reversed.
- An oval or eye-shaped symbol serves as a cue to look further.
- A square means observation modifies the narrative.
- A circle means narrative is supplemented by observation.



Ch6 Agile modelling

Agile methods are a collection of innovative, user-centered approaches to systems development.

Values and Principles of Agile Modelling

- Communication
 - Simplicity
 - Feedback
 - Courage
- Agile modelers possess the self confidence to allow their customers to question, critique, and sometimes complain about the system under development. Analysts learn from their customers, who have been in business a long time.

The Basic Principles of Agile Modelling

- Satisfy the customer through delivery of working software.
- Embrace change, even if introduced late in development.
- Continue to deliver functioning software incrementally and frequently.
- Encourage customers and analysts to work together daily.
- Trust motivated individuals to get the job done.
- Promote face-to-face conversation
- Concentrate on getting software to work.
- Encourage continuous, regular, and sustainable development.
- Adopt agility with attention to mindful design.
- Support self-organizing teams
- Provide rapid feedback
- Encourage quality
- Review and adjust behavior occasionally
- Adopt simplicity.

Four Basic Activities of Agile Modelling: Coding, Testing, Listening, Designing.

Scrum

- Begin the project with a high-level plan that can be changed on the fly.
- Success of the project is most important
- Individual success is secondary
- Project leader has some (not much) influence on the detail
- System team works within a strict time frame (two to four weeks for development)

Scrum is a high-intensity methodology. It is just one of the approaches that adopts the philosophy of agile modelling.

Product backlog — in which a list is derived from product specifications

Sprint backlog — a dynamically changing list of tasks to be completed in the next sprint.

Sprint — a 30-day period in which the development team transforms the backlog into software that can be demonstrated.

Daily scrum — a brief meeting in which communication is the number-one rule

Demo — working software that can be demonstrated to the customer.

Ch7 Data Flow Diagrams

8.7. Graphically characterize data processes and flows in a business system.

- Depict: system inputs, Processes, Outputs

Logical and Physical Data Flow Diagrams

- Logical
 - Focuses on the business and how the business operates.
 - Not concerned with how the system will be constructed
 - Describes the business events that take place and the data required and produced by each event.
- Physical
 - show how the system will be implemented
 - Depicts the system

The Progression of Models from Logical to Physical: Current Logical Data Flow Diagram → New Logical Data Flow Diagram → New Physical Data Flow Diagram.

The Data Dictionary: • a reference work of data about data (metadata); • Collects and coordinates data terms, and confirms what each term means to different people in the organization.

Logic of Decisions.

- Documenting and analyzing logic: ~~structured~~ structured English; Decision tables, Decision trees.
- Logic and structured decisions are distinguishable from semistructured decisions
- Structured decision analysis methods promote completeness, accuracy, and communication.

Ch10 The Unified Modelling Language (UML) Concepts and Diagrams

Things — the objects

Relationships — the glue that holds things together

Diagrams — categorized as either structural or behavioral

UML provides a standardized set of tools to document the analysis and design of a software system

Ch13 Entities. Any object or event about which someone chooses to collect data; May be a person, place or thing; May be an event or unit of time.

Relationships: one-to-one, one-to-many, many-to-many

• A single vertical line represents one

• A crow's foot represents many

Relationships are associations between entities.

Self-join relationship — an entity with a relationship connecting to itself.

Attributes, Records, and Keys

① Attributes represent some characteristic of an entity.

② Records are data items in a record used to identify the record.

Keys
② Records are a collection of data items that have something in common with the entity described.

First Normal Form: Remove repeating groups; The primary key with repeating group attributes are moved into a new table; When a relation contains no repeating groups, it is in first normal form (1NF)

2NF: Remove any partially dependent attributes and place them in another relation; A partial dependency is when the data are dependent on a part of a primary key; A relation is created for the data that are only dependent on part of the key and another data that are dependent on both parts.

3NF: Must be in 2NF; Remove any transitive dependencies; A transitive dependency is when nonkey attributes are dependent not only on the primary key, but also on a nonkey attribute.

